

COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 09

Combinational Circuit



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Recap of Previous Lecture

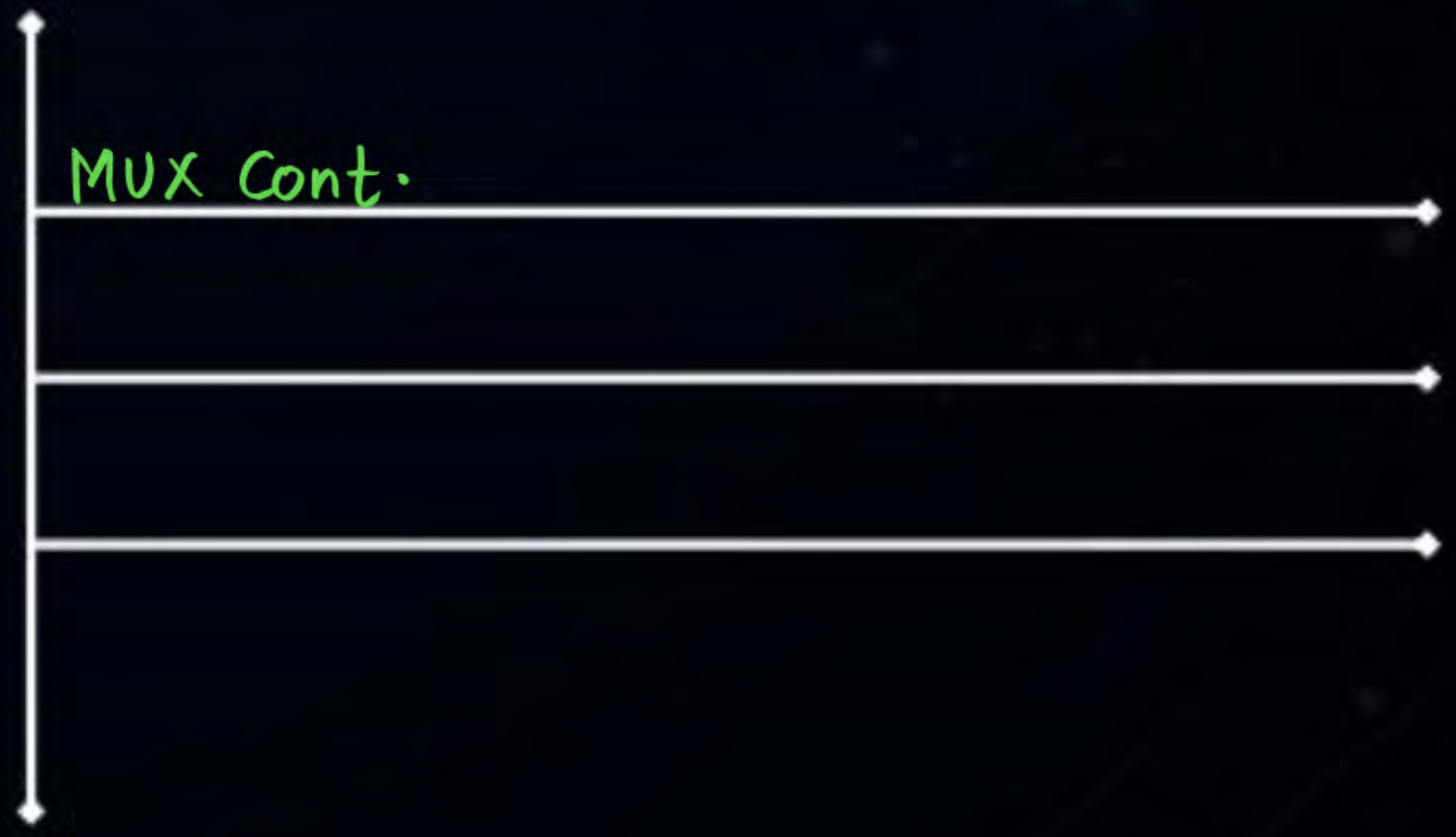
MUX





Topics to be Covered

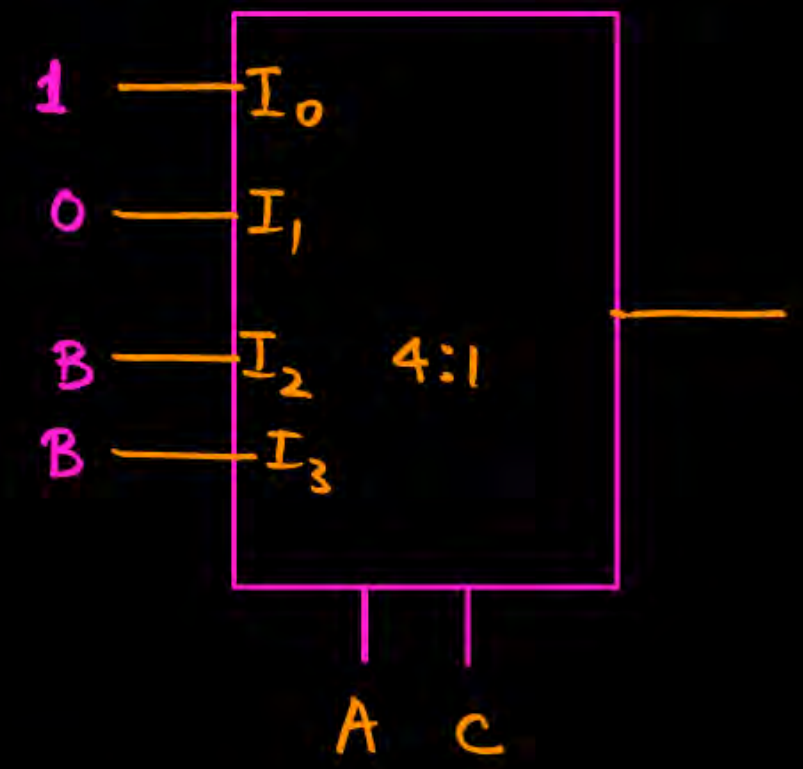
MUX Cont.





• $f(A, B, C) = \Sigma(0, 2, 6, 7)$

using 4:1 MUX & A & C on select line.



	I_0	I_1	I_2	I_3
$\overline{B}$	0	1	4	5
B	2	3	6	7
	1	0	B	B

- 4 : 1 MUX can be used to implement

All functions of 2-variables

Some of the 3-variable function

- 4 : 1 MUX + 1 NOT GATE can be used to implement
└→ All 3-variable functions.

- 8 : 1 MUX can be used to implement

All 3-variable function

Some of the 4-variable function.

- 8 : 1 MUX + 1 NOT GATE can be used to implement

└→ All 4-variable function

- $2^n : 1$ MUX can be used to implement

All n -Variable function

Some of the $(n+1)$ Variable function

- $2^n : 1$ MUX + 1 NOT GATE can be used to implement
 ↳ All of the $(n+1)$ Variable function.

To implement all of the 5-variable function we require

$$2^5 : 1 \text{ MUX}$$

$$2^4 : 1 \text{ MUX} + 1 \text{ NOT}$$

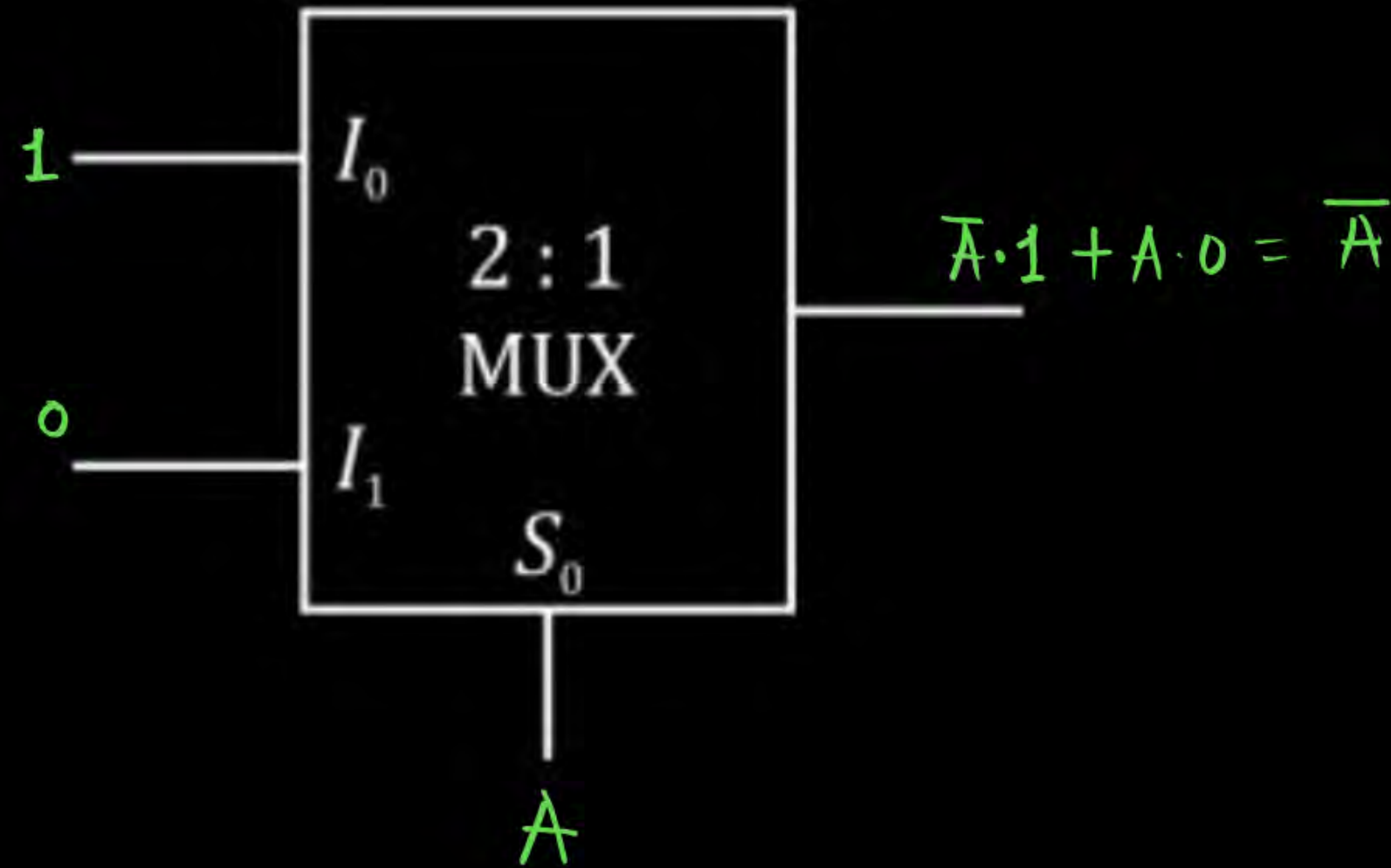
To implement all the functions of $(n+2)$ Variable, we require minimum order MUX (if not got is available $2^{n+1} : 1$).

[MUX as Universal Circuit]

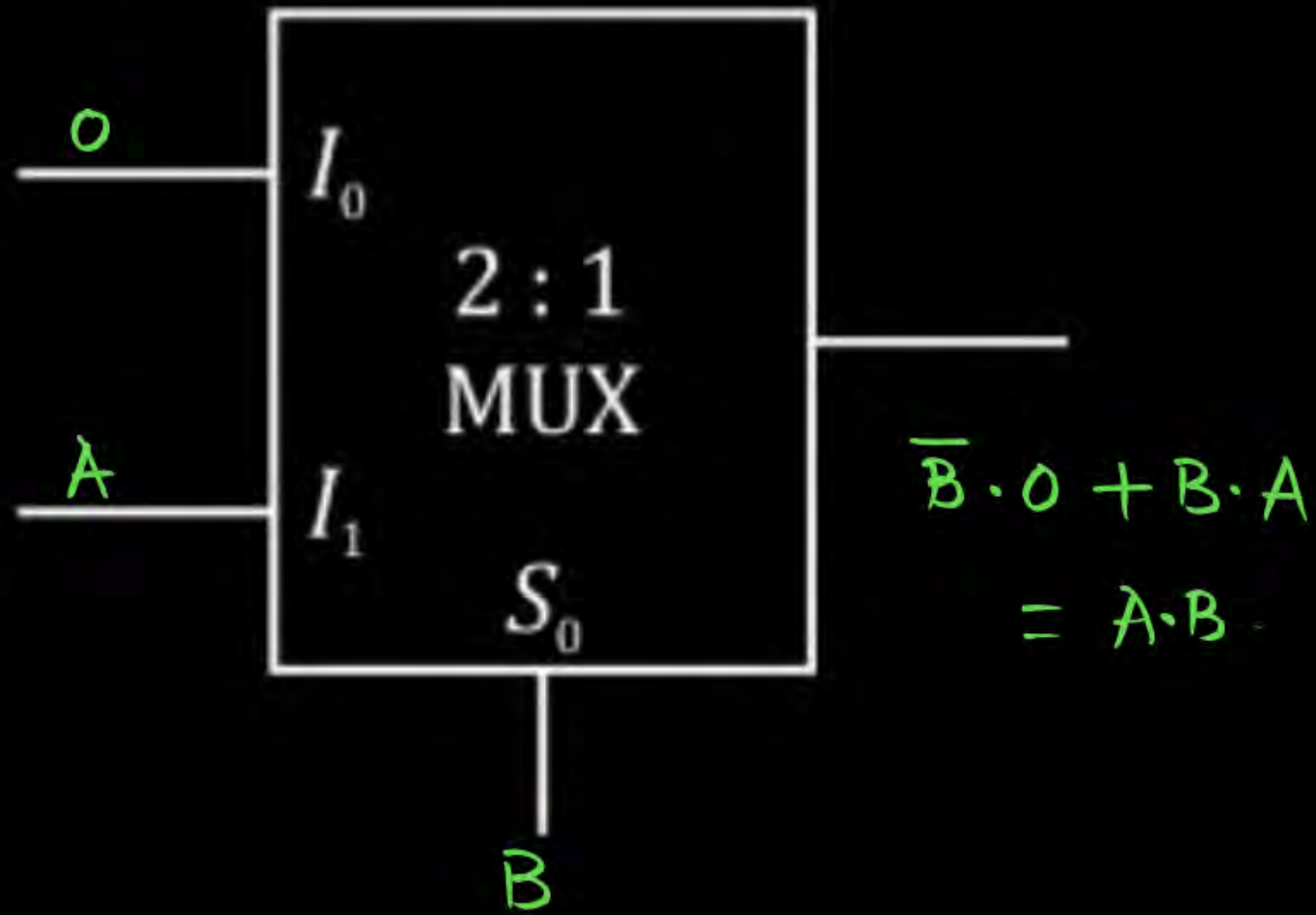
- NOT GATE : $f(A) = \bar{A} = \sum(0)$

$$, f(0) = 1, I_0 = 1$$

$$f(1) = 0, I_1 = 0$$



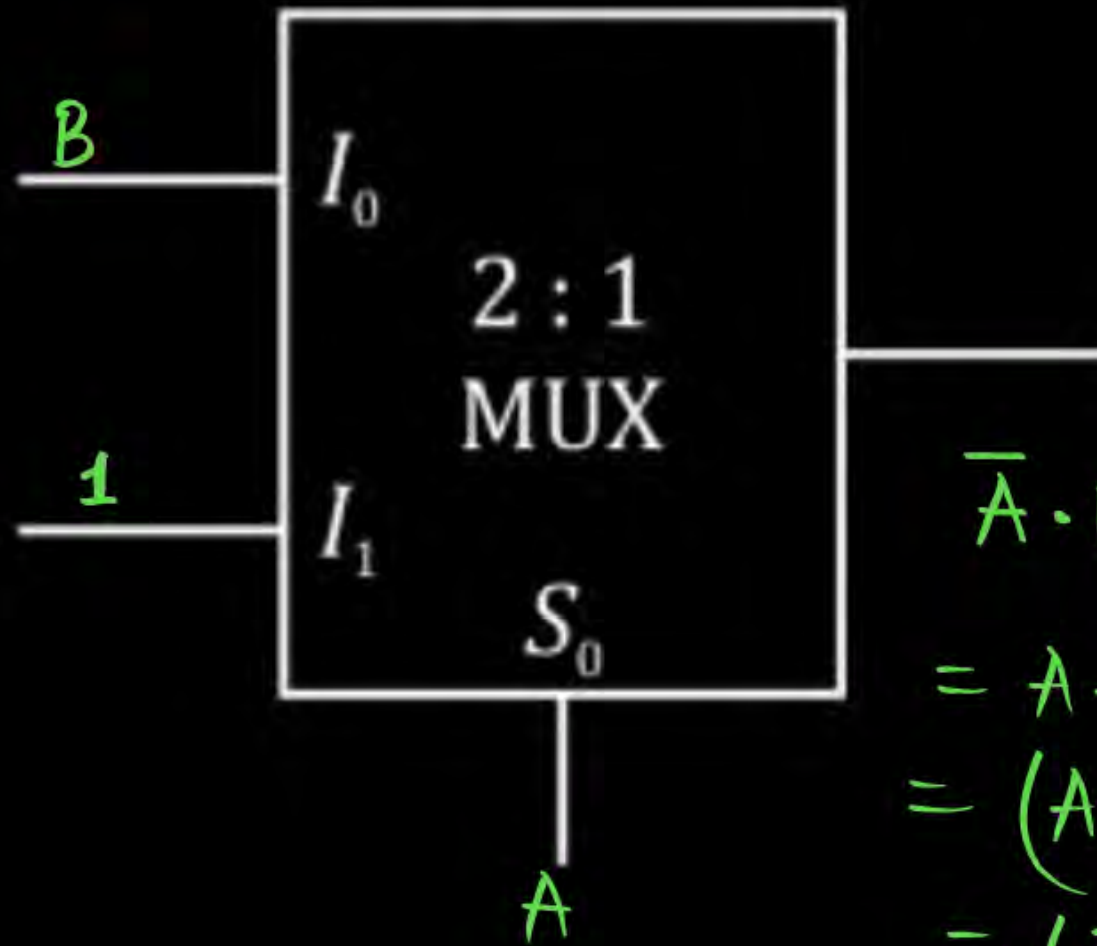
- AND GATE : $f(A, B) = A \cdot B = \Sigma(3)$



$$f(A, 0) = 0, \quad I_0 = 0$$

$$f(A, 1) = A \cdot 1 = A, \quad I_1 = A$$

- OR GATE : $f(A, B) = (A + B) = \Sigma(1, 2, 3)$



$$f(0, B) = B, \quad I_0 = B$$

$$f(1, B) = 1, \quad I_1 = 1$$

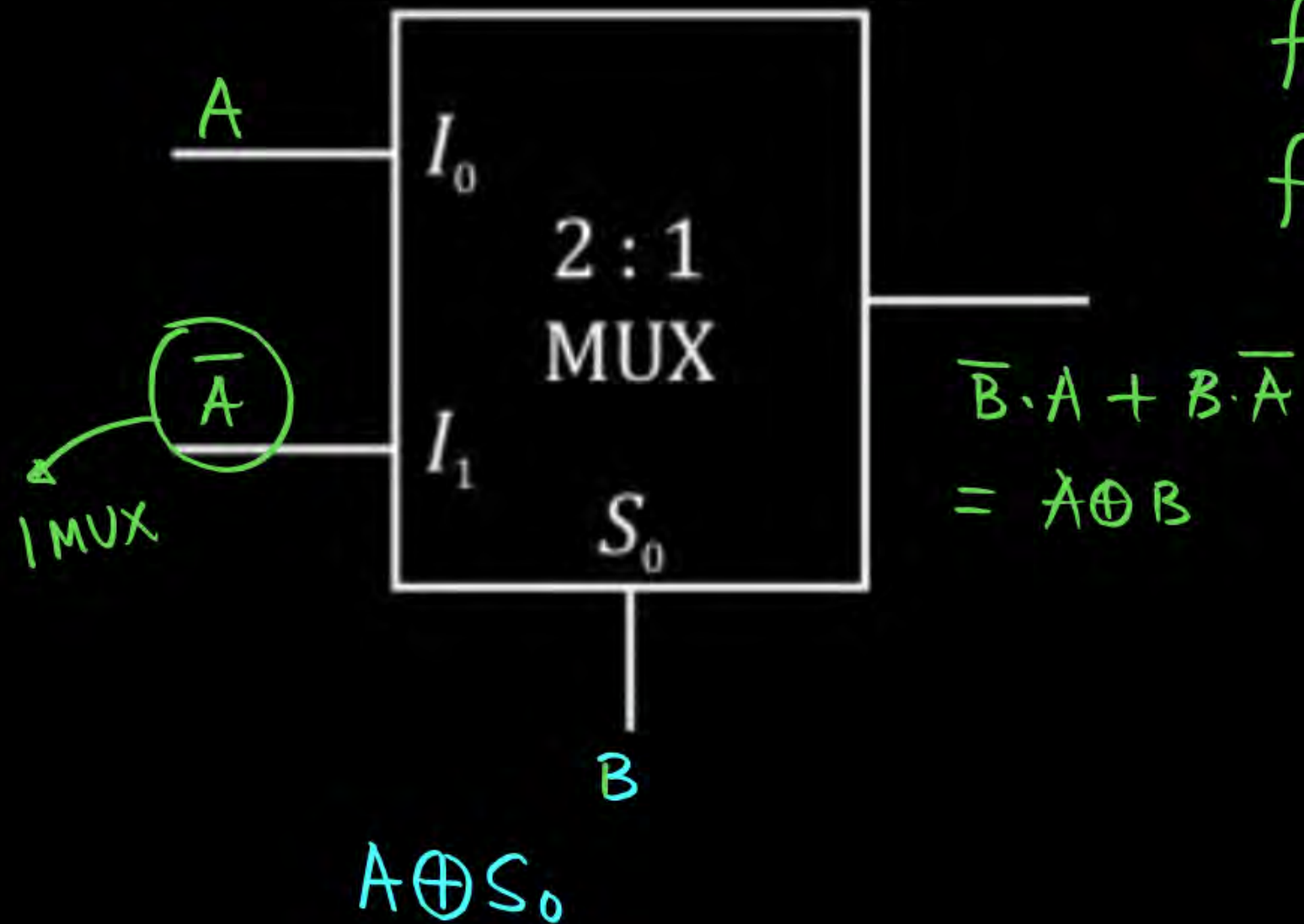
$$\bar{A} \cdot B + A \cdot 1$$

$$= A + \bar{A}B$$

$$= (A + \bar{A})(A + B)$$

$$= (A + B)$$

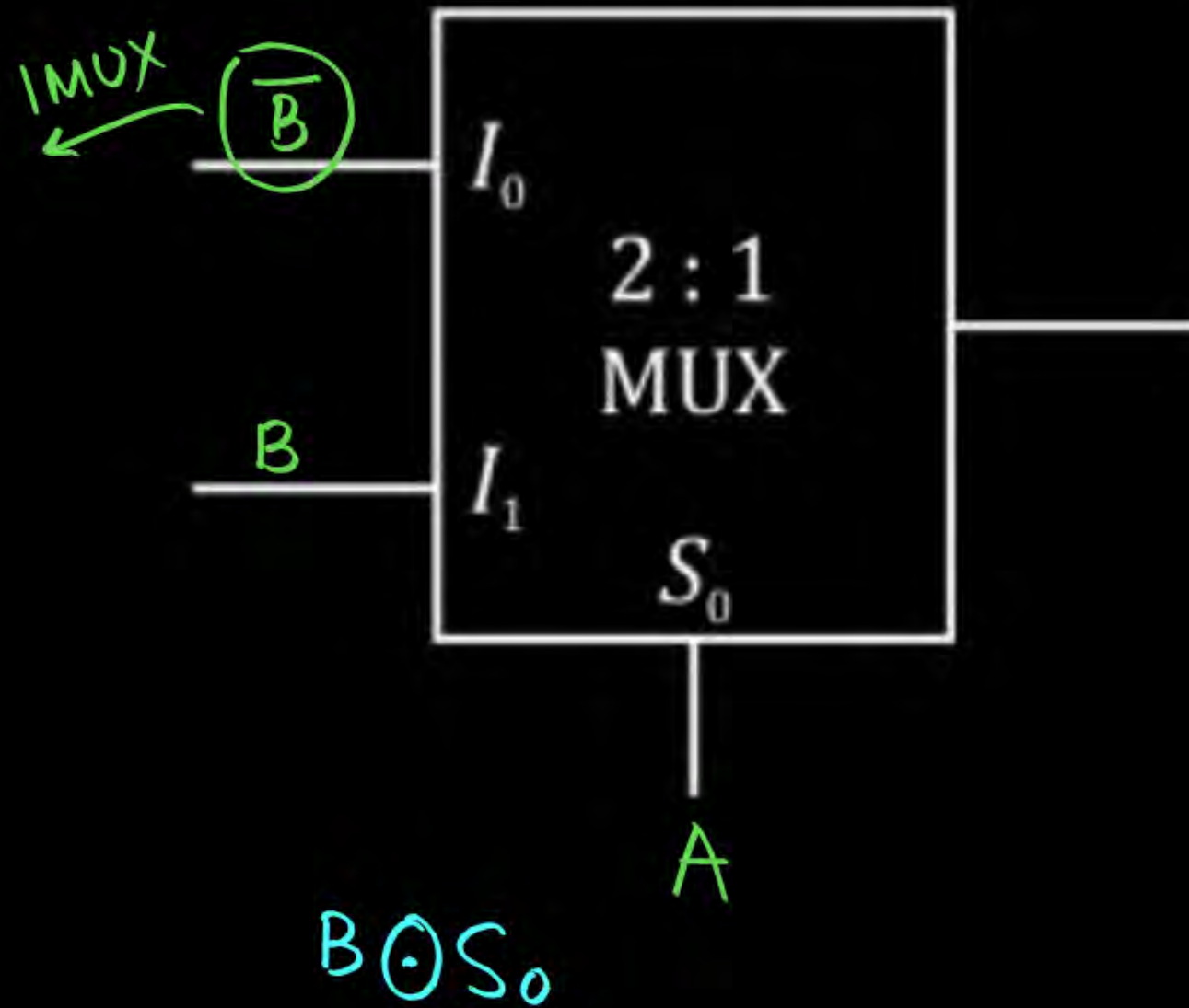
- XOR GATE : $f(A, B) = (A \oplus B) = \Sigma(1, 2)$



$$f(A, 0) = A \oplus 0 = A, \quad I_0 = A$$

$$f(A, 1) = A \oplus 1 = \overline{A}, \quad I_1 = \overline{A}$$

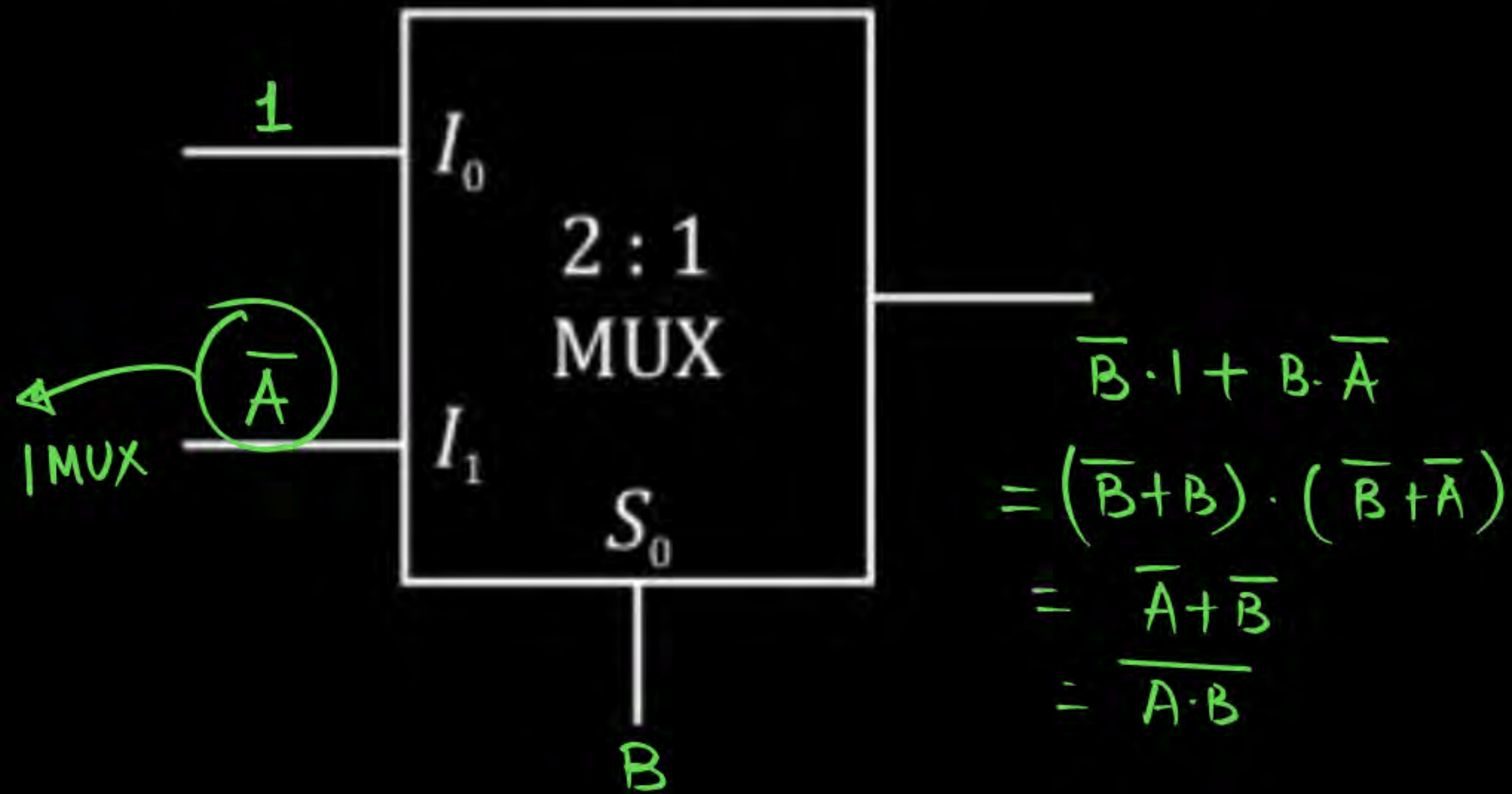
- XNOR GATE : $f(A, B) = (A \odot B)$



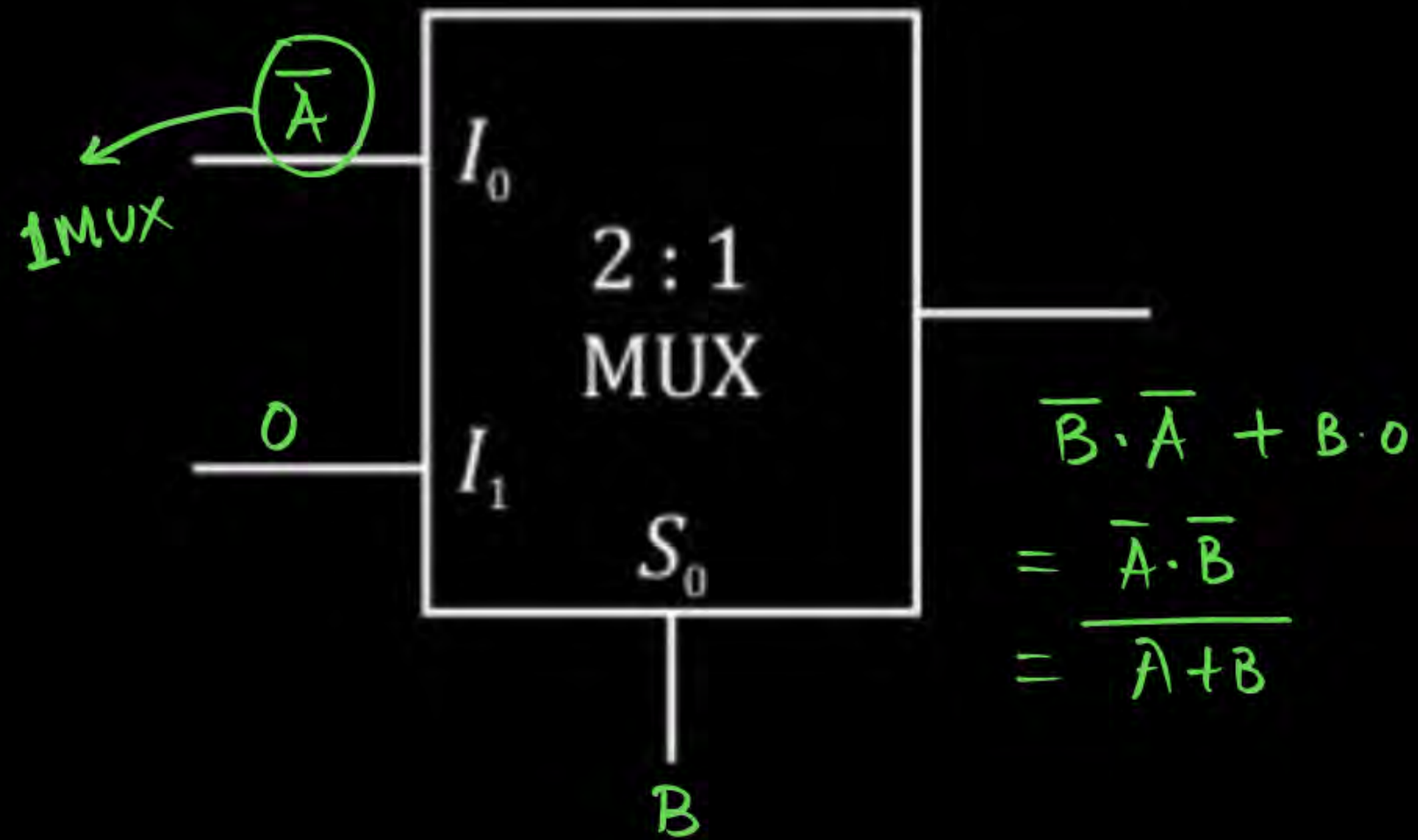
$$f(0, B) = 0 \odot B = \overline{B}, \quad I_0 = \overline{B}$$

$$f(1, B) = 1 \odot B = B, \quad I_1 = B$$

- NAND GATE : $f(A, B) = \overline{A \cdot B} = \Sigma(0, 1, 2)$



- NOR GATE : $f(A, B) = \overline{A + B} = \Sigma(0)$



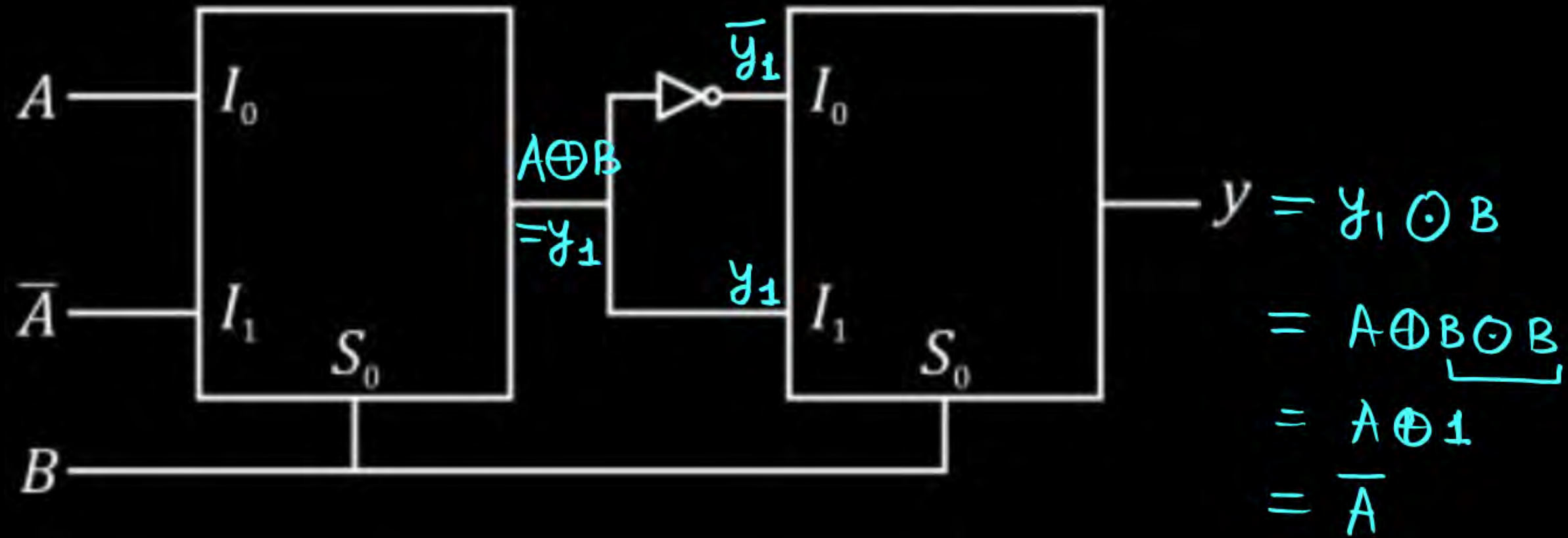
- IMP Notes :

- To implement basic gates [NOT, AND, OR] we require one [2:1] MUX for each.
- To implement non-basic gates [XOR, XNOR, NAND, NOR] we require two [2:1] MUX for each.

[Questions On MUX]



A digital circuit is as given below :



Output y is

(a) $A \oplus B$

(b) $A \odot B$

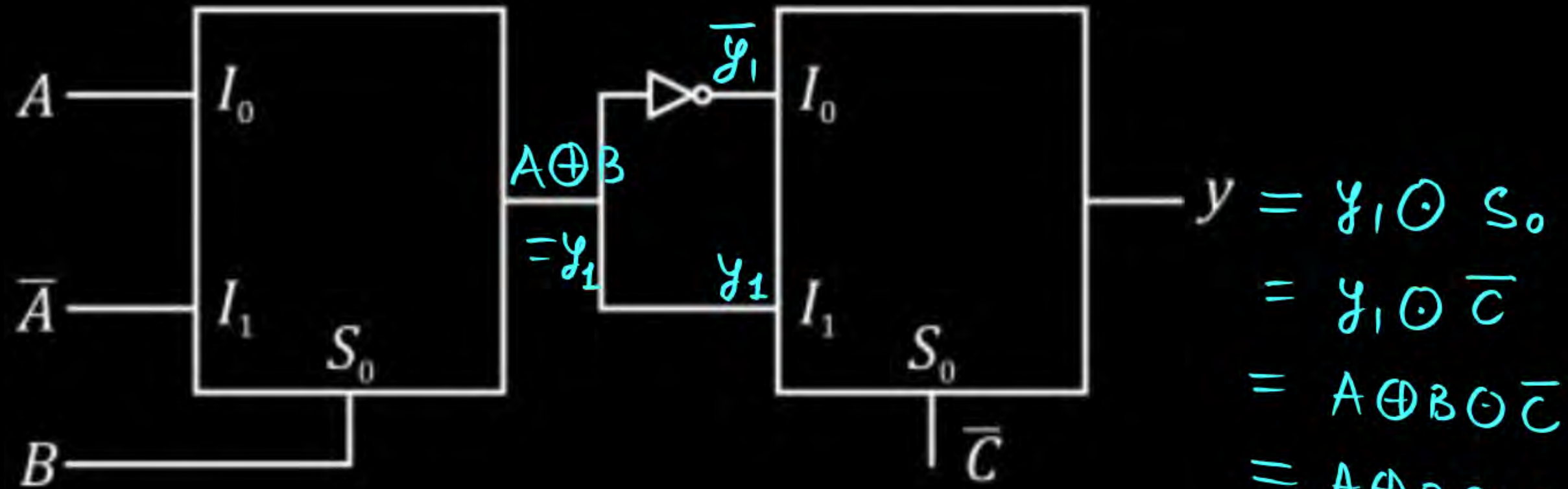
(c) $C.A$

☒ (d) $\bar{A}$

[Question 2]



A digital circuit is as given below :



$$\begin{aligned} y &= y_1 \odot S_0 \\ &= y_1 \odot \bar{C} \\ &= A \oplus B \odot \bar{C} \\ &= A \oplus B \oplus C \\ &= A \odot B \odot C \\ &= \underline{A \oplus B \oplus C} \end{aligned}$$

Output y is

(a) ✓ $A \oplus B \oplus C$

(b) ✗ $A \oplus B \odot C$

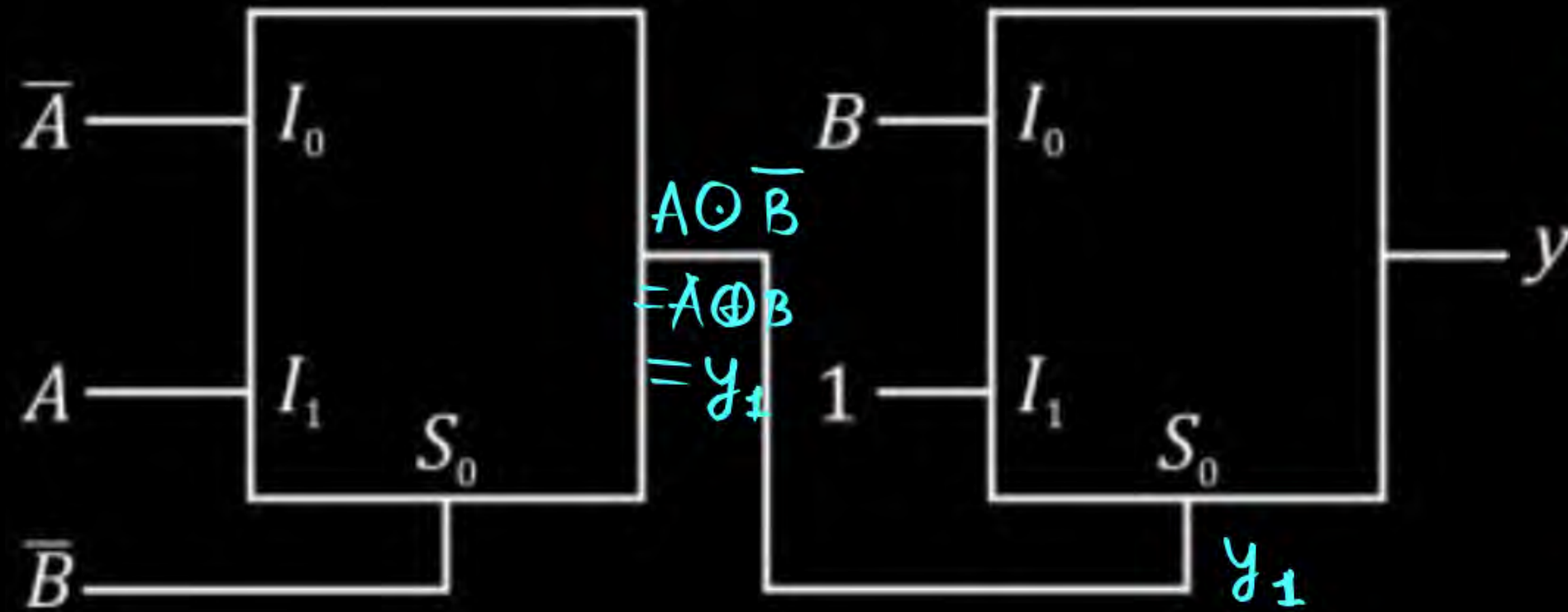
(c) $A \oplus C$ ✗

(d) ✗ $A \oplus B$

Question 3

A digital circuit is as given below :

	$\bar{B}$	B
$\bar{A}$		1
A	1	1



Output y is

(a) $A \oplus B$

(b) $A \odot B$

☒ (c) $(A + B)$

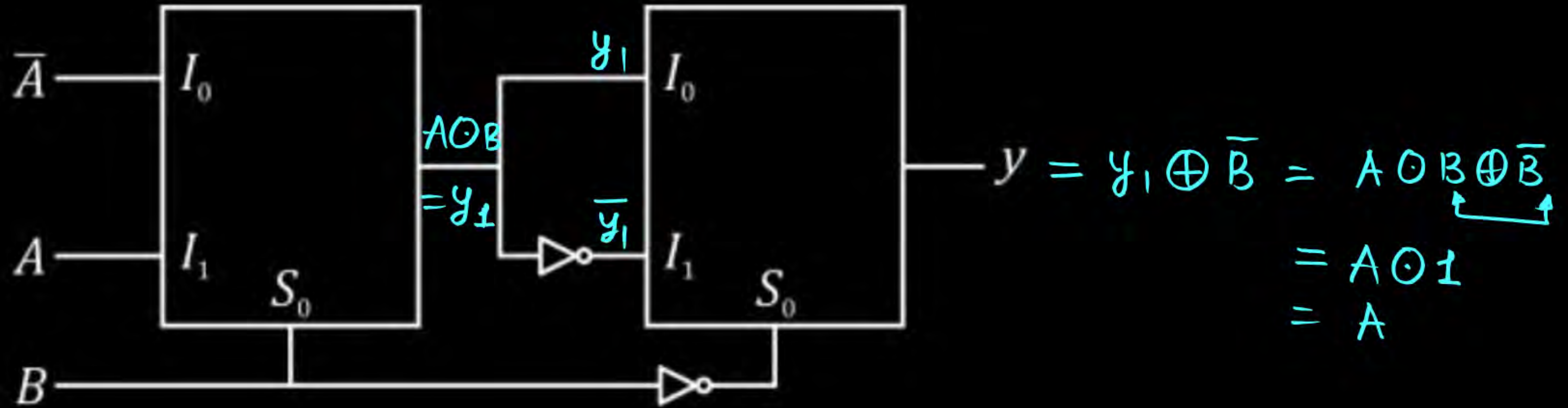
(d) $A \cdot B$

$$\begin{aligned} &= (\bar{A}\bar{B} + AB) \cdot B + \bar{A}B + A\bar{B} \\ &= AB + \bar{A}B + A\bar{B} \\ &= y_1 \cdot B + y_1 \cdot 1 \\ &= y_1 + B \\ &= A \oplus B + B \\ &= \bar{A}B + A\bar{B} + B \\ &= B + A\bar{B} \\ &= (B + A) \end{aligned}$$

[Question 4]



A digital circuit is as given below :



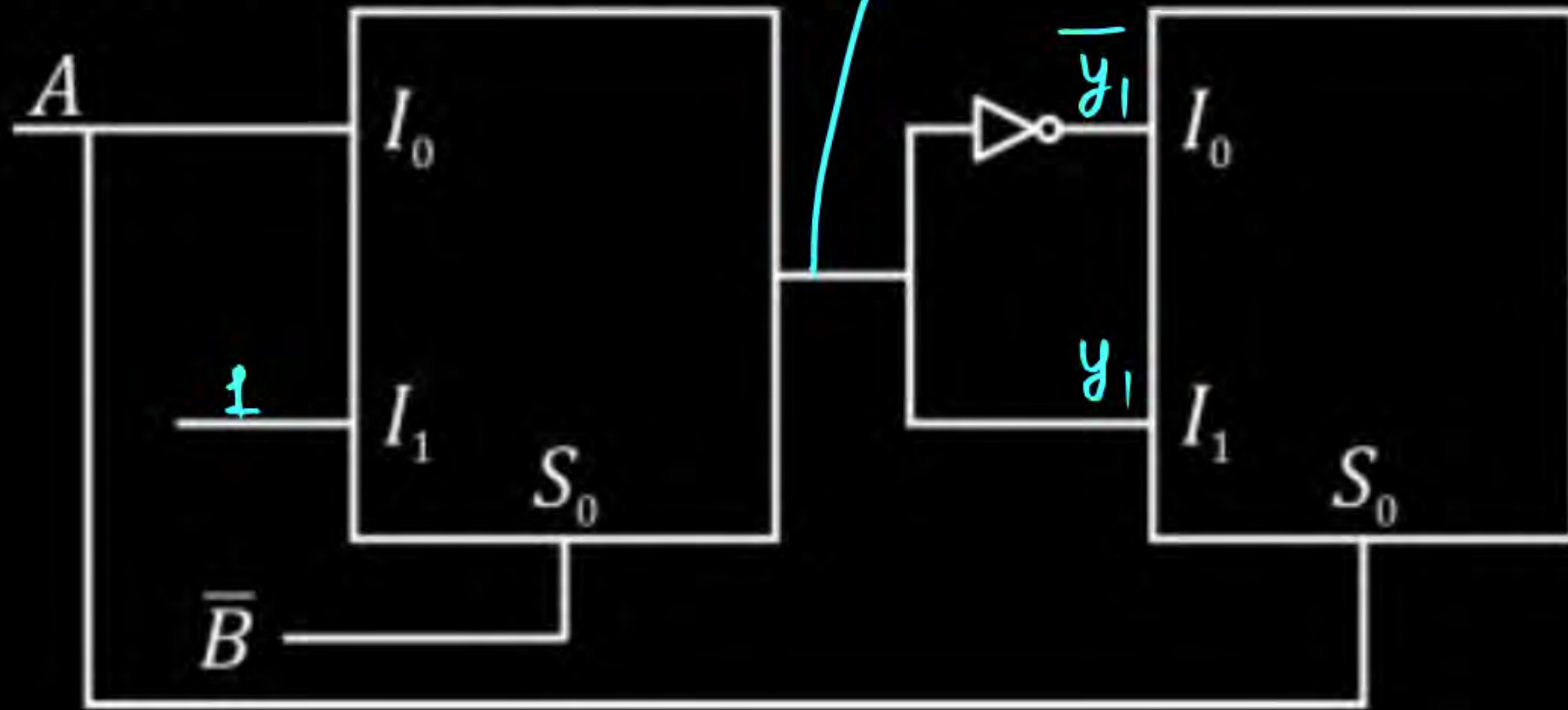
Output y is

- (a) B
- (b) A ✓
- (c) $(A + B)$
- (d) $A . B$

[Question 5]



A digital circuit is as given below.



$$\begin{aligned} y_1 &= \bar{S}_0 \cdot A + S_0 \cdot 1 = B \cdot A + \bar{B} \\ &= (A + \bar{B}) \\ &= \bar{B} + B \cdot A \\ &= (\bar{B} + B) \cdot (\bar{B} + A) \end{aligned}$$

$$\begin{aligned} y &= y_1 \odot A \\ &= (A + \bar{B}) \odot A \\ &= \bar{A} \cdot B \cdot \bar{A} + (A + \bar{B}) \cdot A \\ &= A + \bar{A} \cdot B \\ &= (A + \bar{A}) (A + B) \\ &= (A + B) \end{aligned}$$

The output y is

(a) $A \odot B$

(b) $\bar{A}B$

(c) $(A + B)$

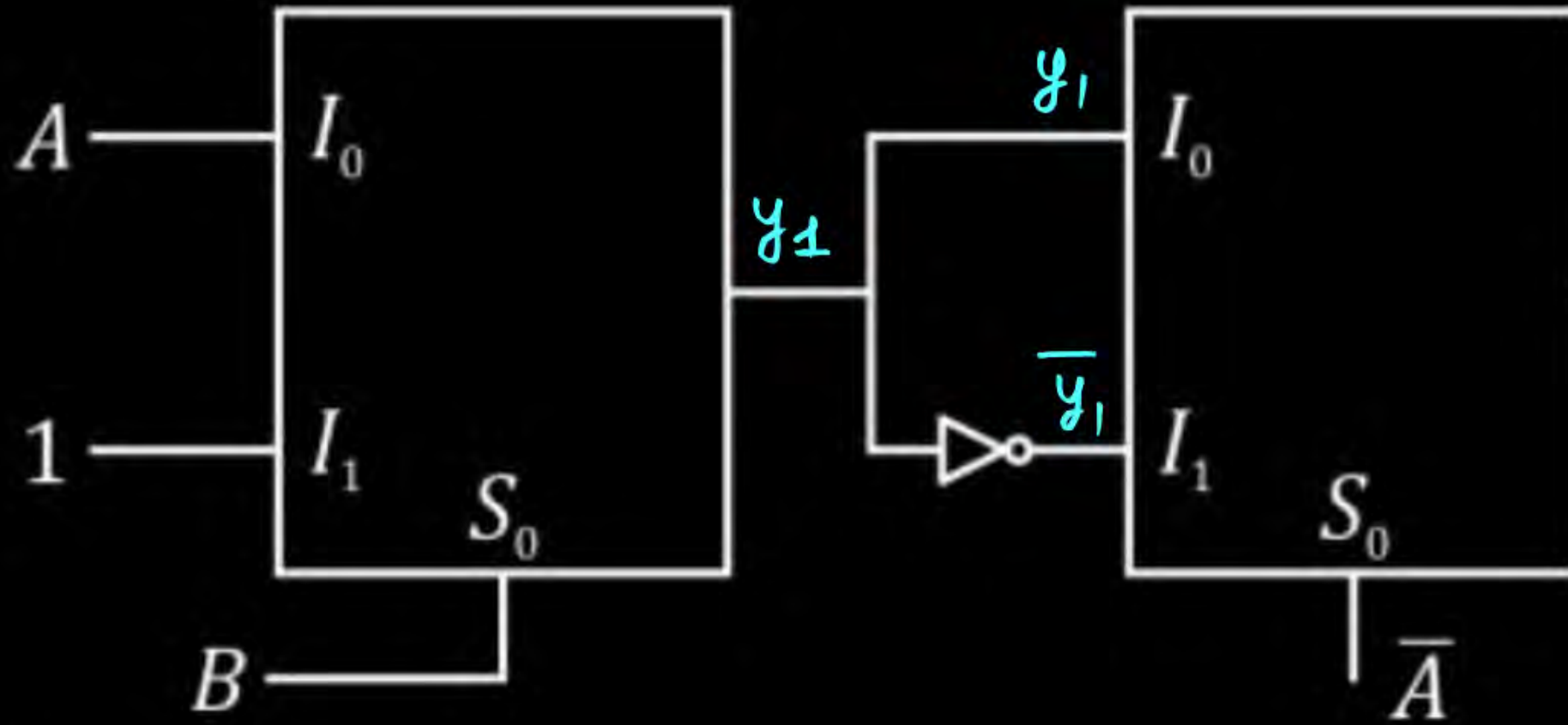
(d) $A\bar{B}$

[Question 6]

$$y_1 = \overline{B} \cdot A + B \cdot 1 = (A + B)$$



A digital circuit is as given below :



Output y is

(a) ~~1~~

~~(b) $A \odot B$~~

☒ (c) $\overline{A}B = A + \overline{B}$

(d) $\overline{A \cdot B}$

$$\begin{aligned} y &= y_1 \oplus \overline{A} \\ &= (A + B) \oplus \overline{A} \\ &= \overline{A} \overline{B} \cdot \overline{A} + (A + B) A \\ &= A + \overline{A} \cdot \overline{B} \\ &= (A + \overline{A}) \cdot (A + \overline{B}) \\ &= (A + \overline{B}) \end{aligned}$$

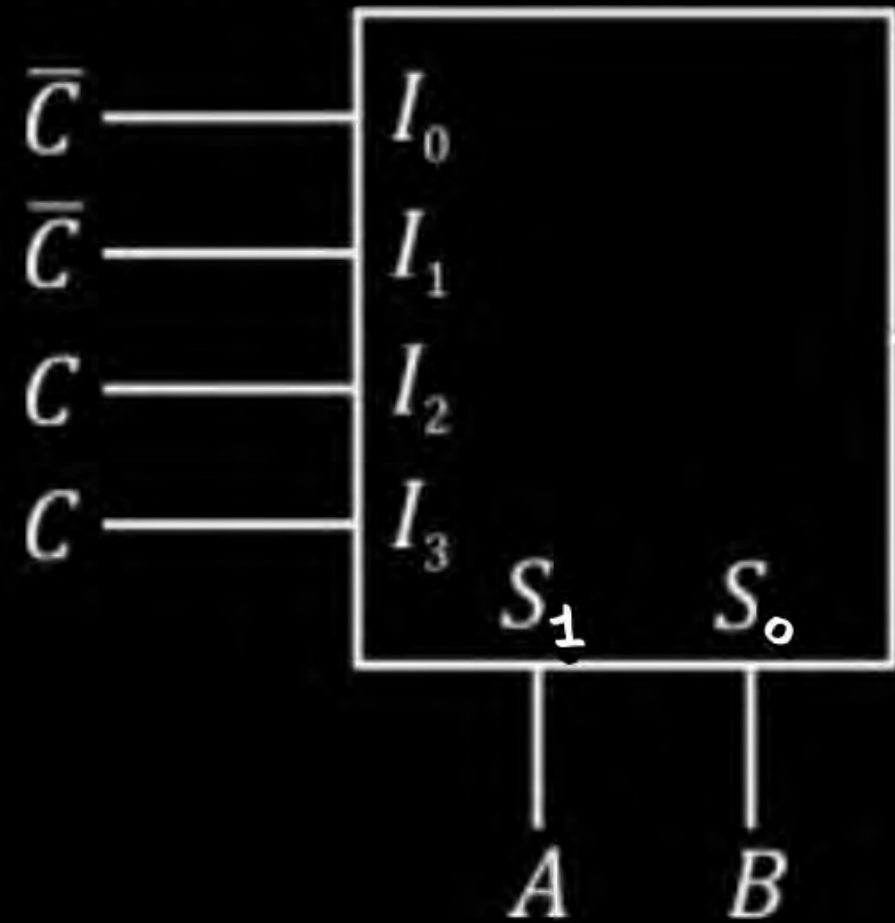
[Question 7]



A digital circuit is as given below :

$$A=0, B=0$$

$$A=0, B=1$$



$$y = \bar{A}\bar{B}.\bar{C} + \bar{A}B.\bar{C} + A\bar{B}.C + ABC$$

$$= \bar{A}\bar{C} + AC$$

$$= A \odot C$$

Output y is

(a) ~~X~~ $A \oplus B \oplus C$ C

(b) ☒ $A \odot C$ $\bar{C}, \bar{C}$

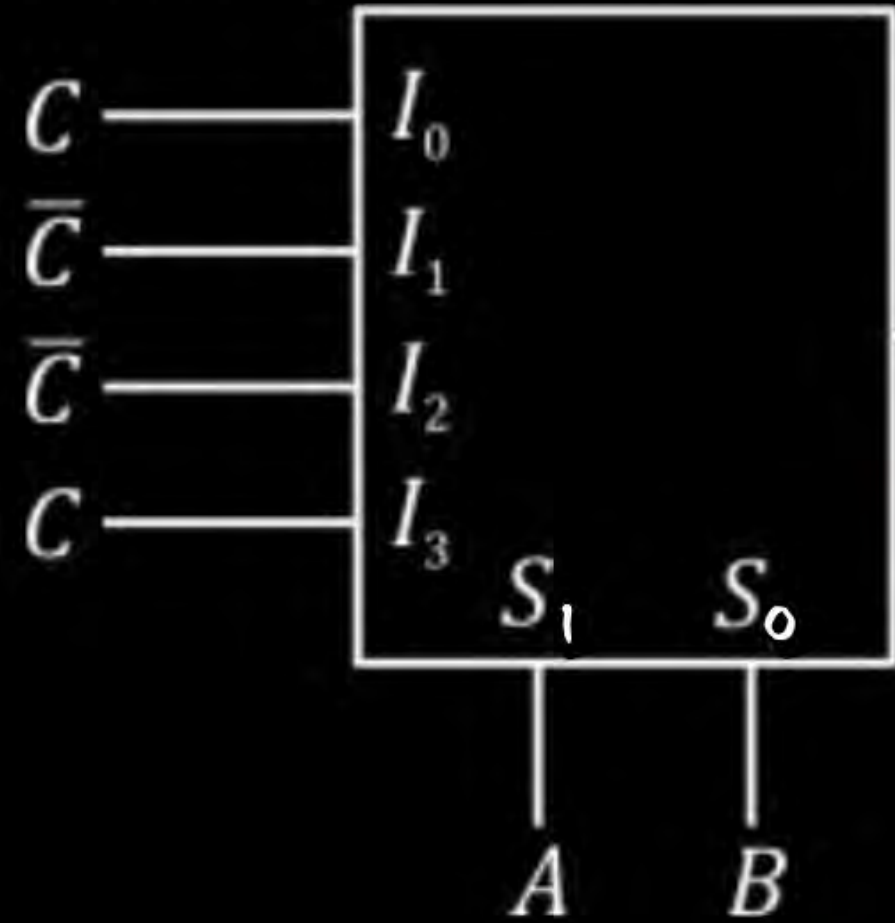
(c) ~~X~~ $A \odot B \oplus C$ $\bar{C}, C$

(d) ~~X~~ $A \oplus B$ 0

[Question 8]

A digital circuit is as given below :

$A=0, B=0$



	$\bar{B}\bar{C}$	$\bar{B}C$	BC	$B\bar{C}$
$\bar{A}$		1		1
A	1		1	

$$\begin{aligned}
 y &= \bar{A}\bar{B} \cdot C + \bar{A}B \cdot \bar{C} + A\bar{B} \cdot \bar{C} + AB \cdot C \\
 &= \Sigma(1, 2, 4, 7) \\
 &= \bar{A} [B \oplus C] + A [B \odot C] \\
 &= \bar{A} y_1 + A \bar{y}_1 \\
 &= A \oplus y_1 = A \oplus B \oplus C \\
 &= \bar{A} \bar{y}_2 + A y_2 = A \odot y_2 \\
 &= A \odot B \odot C
 \end{aligned}$$

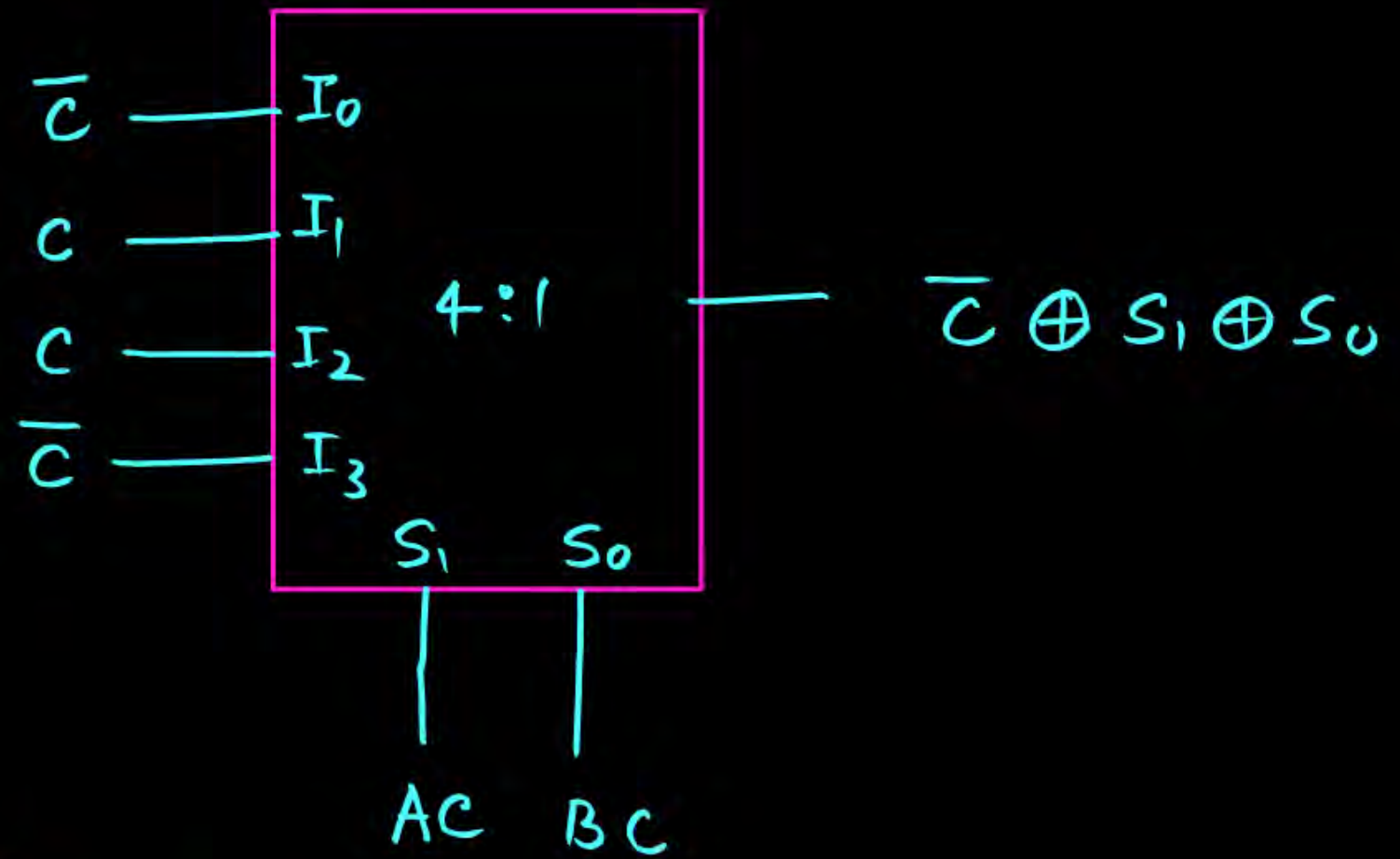
Output y is

(a) $A \oplus B \oplus C$ ✓

(b) $A \odot B \oplus C$ ✗

(c) $\overline{A \oplus B \odot C}$ ✗

(d) $A \oplus B$ ✗

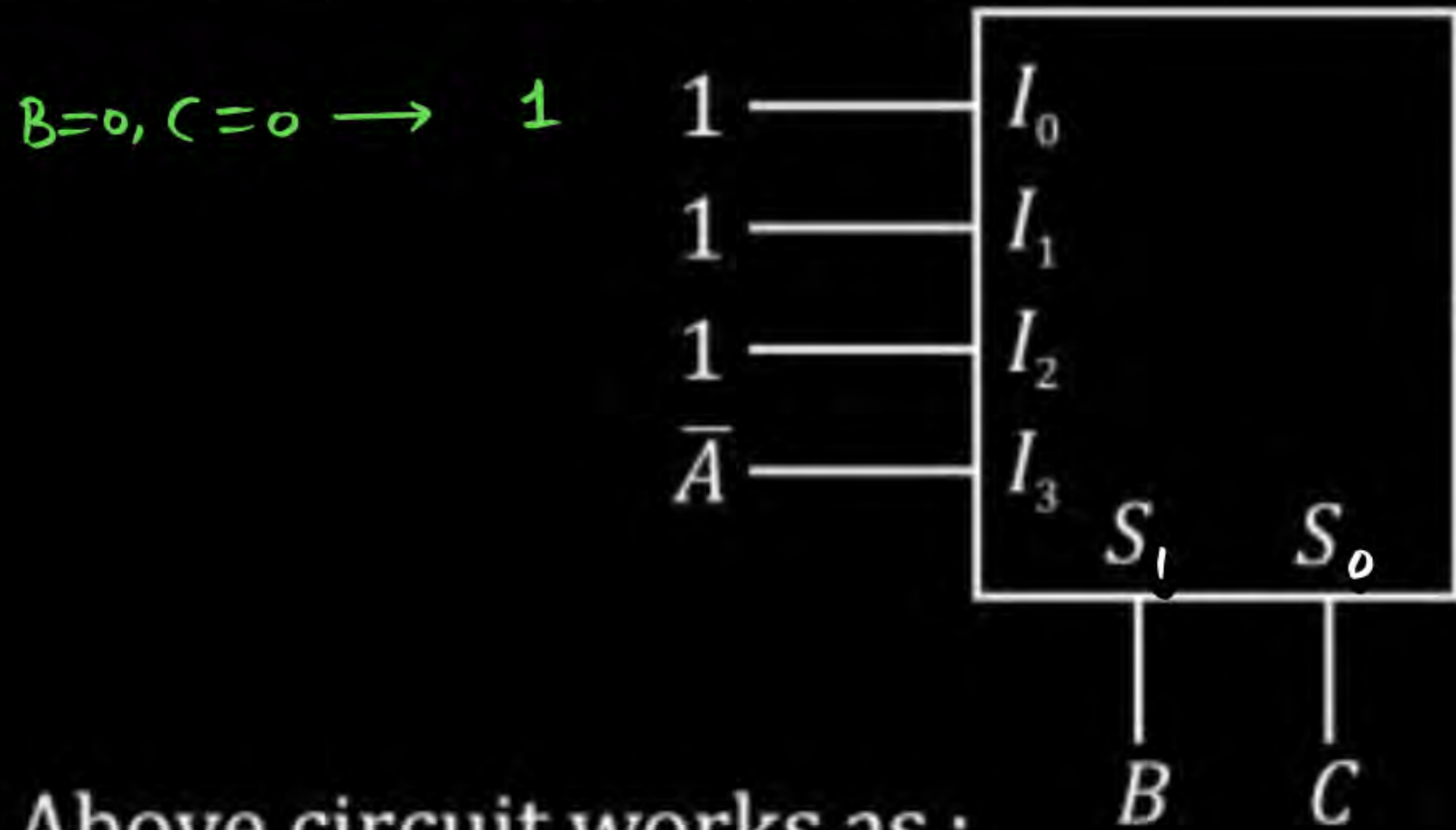


$$\begin{aligned}
 &= \overline{C} \oplus AC \oplus BC \\
 &= \overline{C \oplus AC \oplus BC} \\
 &= C [1 \oplus A \oplus B] \\
 &= C \cdot [A \oplus B]
 \end{aligned}$$

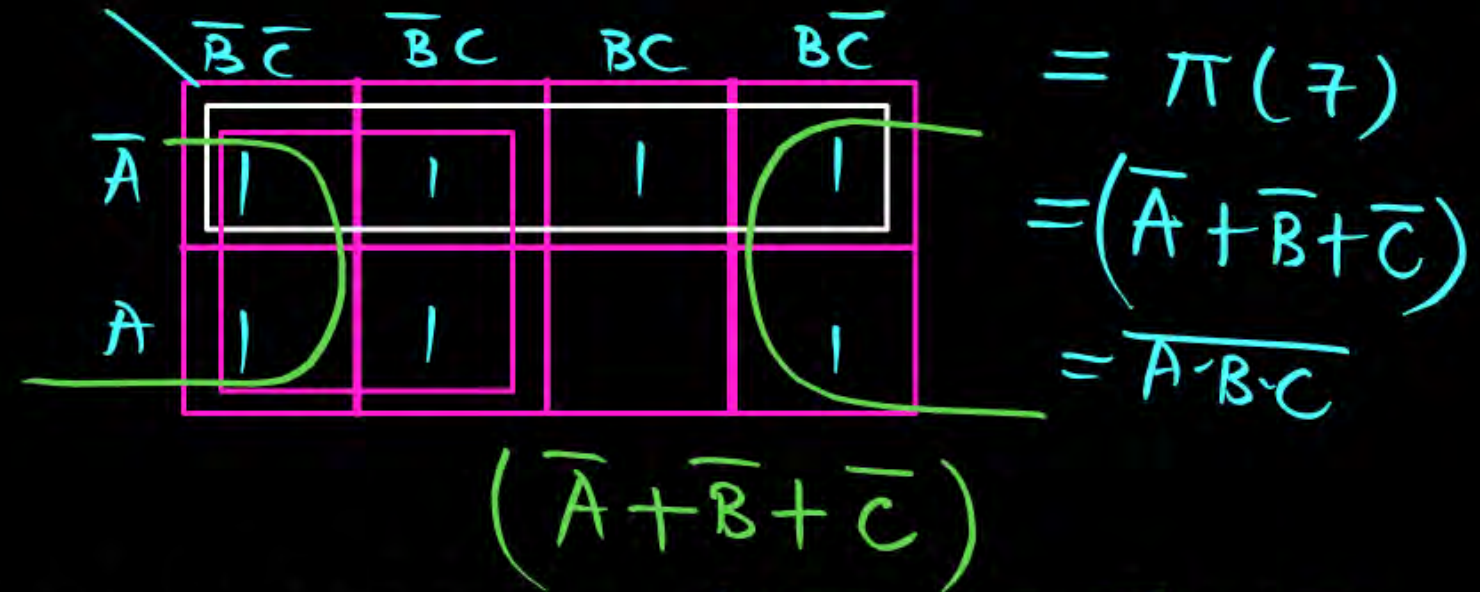
[Question 9]



A digital circuit is as given below :



$$Y = \overline{B}\overline{C} \cdot 1 + \overline{B}C \cdot 1 + B\overline{C} \cdot 1 + BC \cdot \overline{A}$$



Above circuit works as :

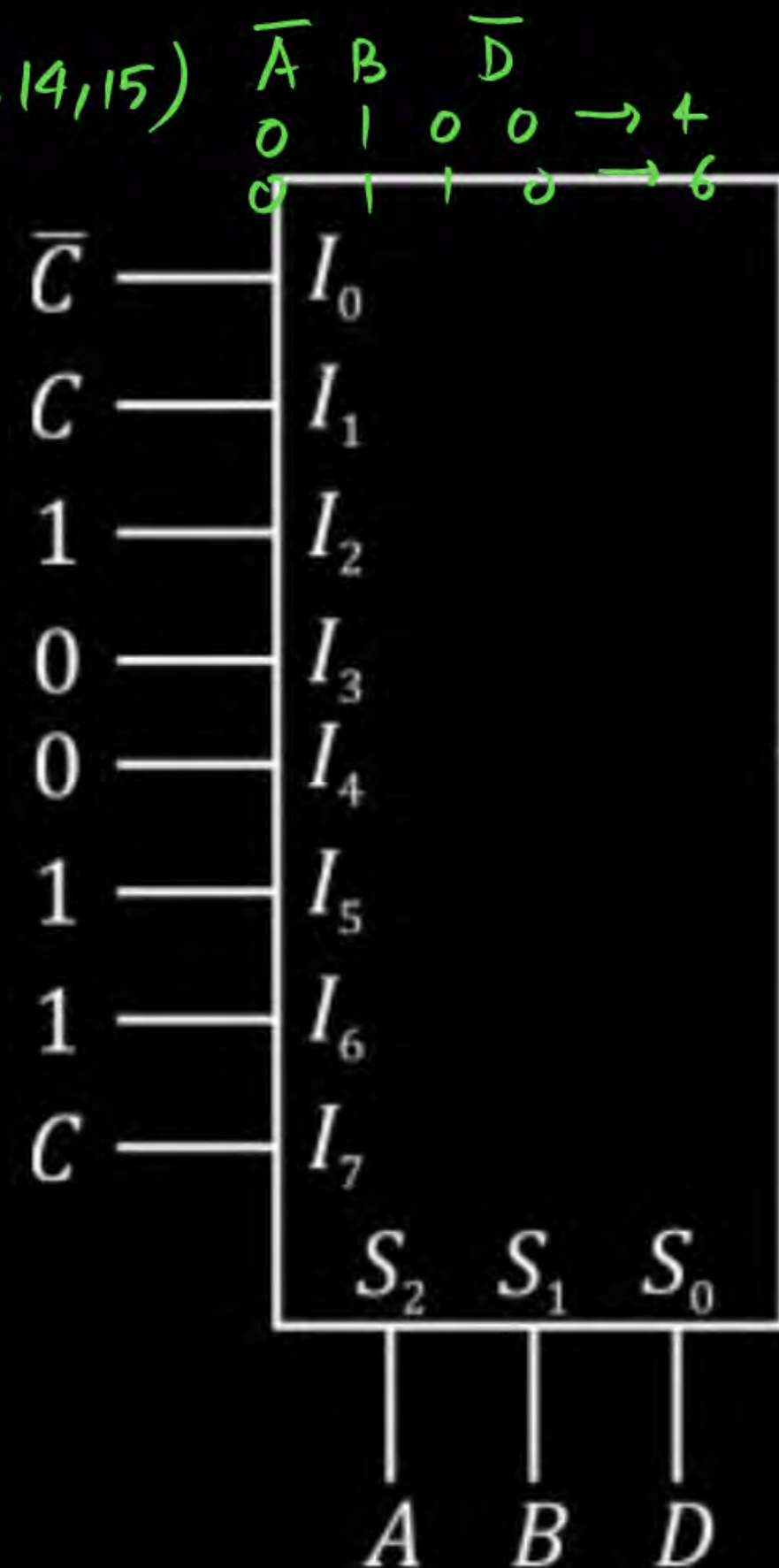
- (a) ~~3-input OR gate~~ A (b) ~~3-input NOR gate~~ $\overline{(A+0+0)} = \overline{A}$
- (c) ~~3-input AND gate~~ $\overline{A \cdot 0 \cdot 0} = 1$ (d) ☒ 3-input NAND gate $\overline{A \cdot 0 \cdot 0} = 1$

Question 10 $\Sigma(0, 3, 4, 6, 9, 11, 12, 14, 15)$

A digital circuit is as given below:

Output y will be: $f(A, B, C, D)$

- (a) $y = \Sigma(0, 3, 4, 6, 7, 9, 11, 12, 14, 15)$ ~~X~~
- (b) $y = \Sigma(0, 1, 3, 4, 6, 8, 9, 11, 12, 14, 15)$ ~~X~~
- (c) $y = \Sigma(0, 3, 4, 6, 9, 11, 12, 14, 15)$ ✓
- (d) $y = \Sigma(0, 3, 4, 6, 9, 10, 11, 12, 14, 15)$ ~~X~~



Handwritten truth table for the output y :

A	$\bar{B}$	D	Value
1	0	0	9
1	0	1	11

Handwritten truth table for the output y :

A	B	$\bar{D}$	Value
1	1	0	12
1	1	1	14

Handwritten Boolean expression for the output y :

$$y = \bar{A}\bar{B}\bar{D}\bar{C} + \bar{A}\bar{B}CD + \bar{A}B\bar{D} + A\bar{B}D + AB\bar{D} + ABCD$$

$$\bar{A} B \quad D \cdot 1$$

$$0 \ 1 \ 0 \ 1 \rightarrow 5$$

$$0 \ 1 \ 1 \ 1 \rightarrow 7$$

$$A \bar{B} \quad \bar{D} \cdot 1$$

$$1 \ 0 \ 0 \ 0 \rightarrow 8$$

$$1 \ 0 \ 1 \ 0 \rightarrow 10$$

Thank you

GW
Soldiers !

